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Part-B

6. HDFS Daemons:-

Daemon is a process that runs in the background. it is mainly done in the Java. there are various daemons present in Hadoop.

i) Name Node

ii) Data Node

iii) Secondary Name Node

iv) Resource Node

v)

what is Hadoop?

Hadoop is a free and open source software used for managing & montoring of bugs amount of data. it can handle structured, unstructured, semi-structured data. Hadoop uses the distributed manner to store the large amount of files. then by Map Reducer it is parallely processing the files & generating the output.

Feature of Hadoop:-

- can handle large amount of data.
- can visualize large amount of data.
- no need of SQL structures i,e No SQL.
- columner way of storing.

Hadoop Daemon:-

→ it schedules, manages the files in the background.

i) Name Node:-

Name Node is mainly responsible to store the Metadata of file. it don't store the actual content of the file. It is the Master Node. it manages the file name, date of modification, last modified.

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in the form of Metadata.

Features of Name node:-

- it donot store the actual data of the file.
- It increases the memory speed & reduces RAM usage.

ii) Data Node:-

Data node is responsible to store the data in HDFS(Hadoop distributed file System). it manages the data present in the file. It is govern by the Name Node. i.e Name Node Instruct the data Node where to store the data.

iii) Secondary Name Node:-

it tells the where the data is present. It is the global Master Node that controls the flow of the Data from Name Node to Data Node. it stores the all the log files in the file called 'fsimage'.

Hadoop Daemon	Port Number
Name Node	50070
Data Node	50075
Secondary Name Node	50090

iv) Resource Node:-

It manages the Resources that are present in the Hadoop ecosystem. there are two types of Resource Managers

a) Resource Manager

b) YARN(Yet Another Resource Nagotiator)

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a) Resource Manager:-

as the name suggests it manages the all resources presenth in the Hadoop esosystem.

b) YARN (Yet another Resource Nagotiator):-

since, in the Hadoop V1 the resource management & the data processing was the part of Map Reduce due to more amount of data we have to get lot of lag in the processing. as a result from next versions the YARN (Yet another Resource Nagotiator) was introduced.

Hadoop Daemon's	Port Number
Resource Node	8080

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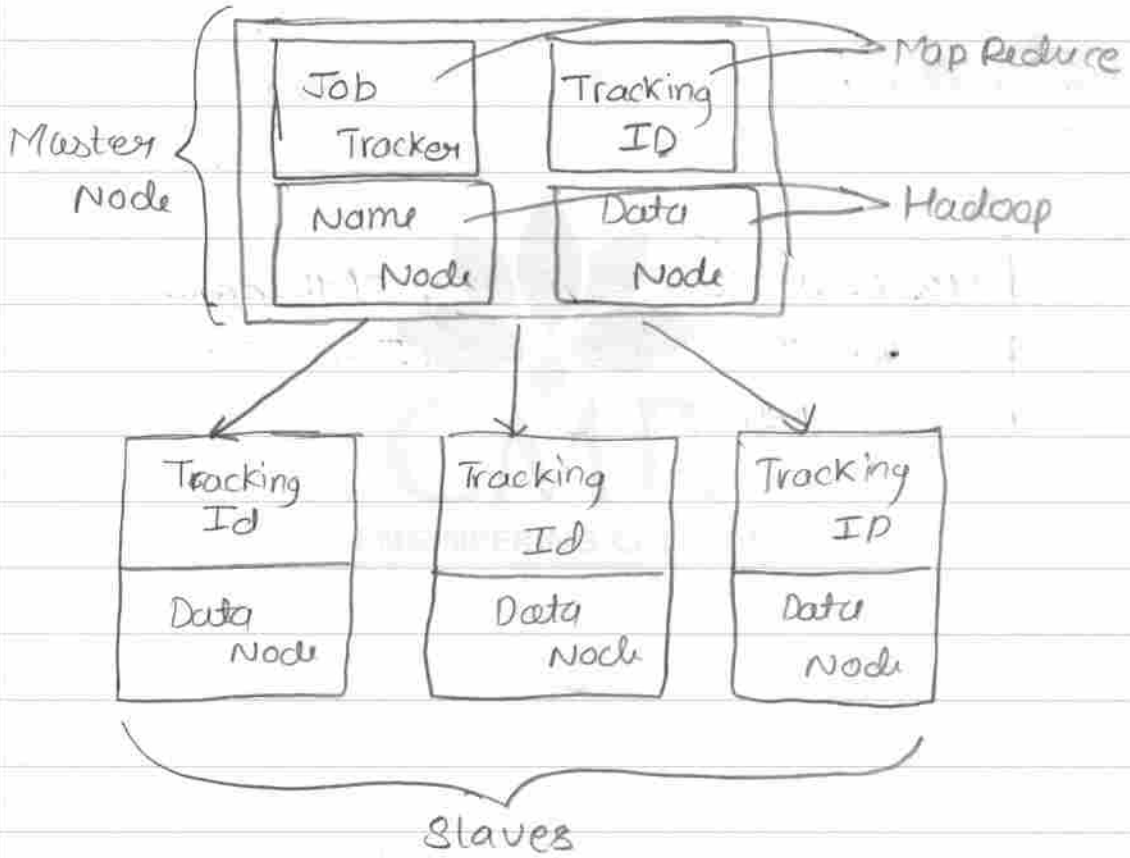
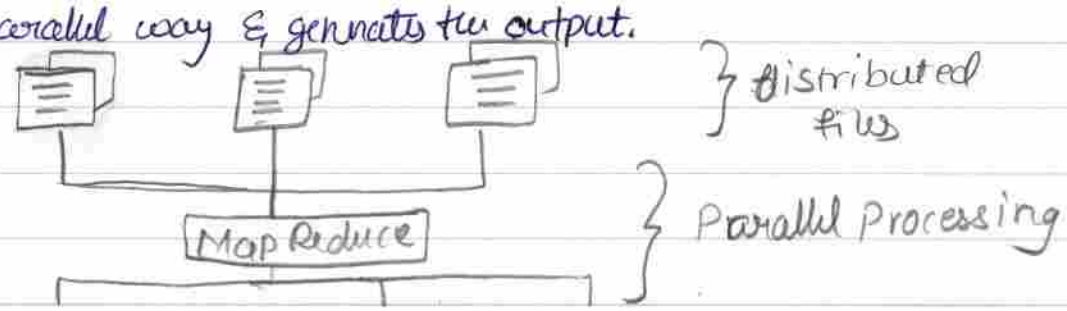
8. Hadoop Distributed File System (HDFS) architecture:-

coming to Hadoop architecture it has got various parts such as Jobtracker, DataNode, NameNode, TrackerID. it is a Master & slave management. where the Master Node has Jobtracker, Data Node, Name Node, TrackerID. It has various slaves working under the Master Node. Below diagram demonstrate the Hadoop distributed File System (HDFS) Architecture.



Map Reduce:- it is a Data Processing unit in the Hadoop ecosystem. which processes the distributed manner files in a parallel way & generates the output.



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Hadoop:- Hadoop is a open source platform that is used to manage the huge amount of data. more over, to be stored in the distributed manner.

JobTracker:- it is used to divide the large number of jobs into smaller jobs & assign it to the tracker id. so that Tracker can give to the respective slave to generate the output.

Name Node:- it is used to maintain the all the slaves & its works that are given. It guids the data node.

DataNode:- it handles the storing of the data in the HDFS & perticular ID given to that slave. it helps in maintaining the data properly.

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10- Machine learning Techniques:-

Machine learning is a term that trains a model to learn from the data given & then predict the future values based on the trained data. we have various types of Machine learning techniques.

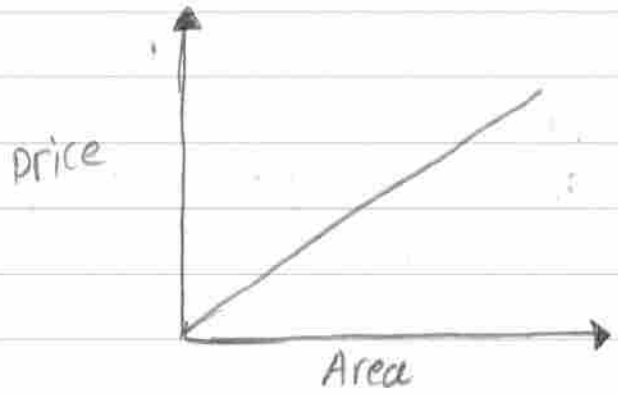
i) Supervised learning
ii) Unsupervised learning
iii) Semi-supervised learning
iv) Reinforcement Learning.

i) Supervised learning:-

a labelled data is given to the model training. label data means the X & Y coordinates. for example let us consider the 'House Price Prediction'. In this we are giving the labelled data as Price & area of the House in sq feets so, the price is the 'Y' as it is the output. & the area is in the input therefore it is 'X'.

Techniques:-

i) linear Regression
ii) K-mean Algorithm.
iii) Multilinear Regression.



i) linear Regression

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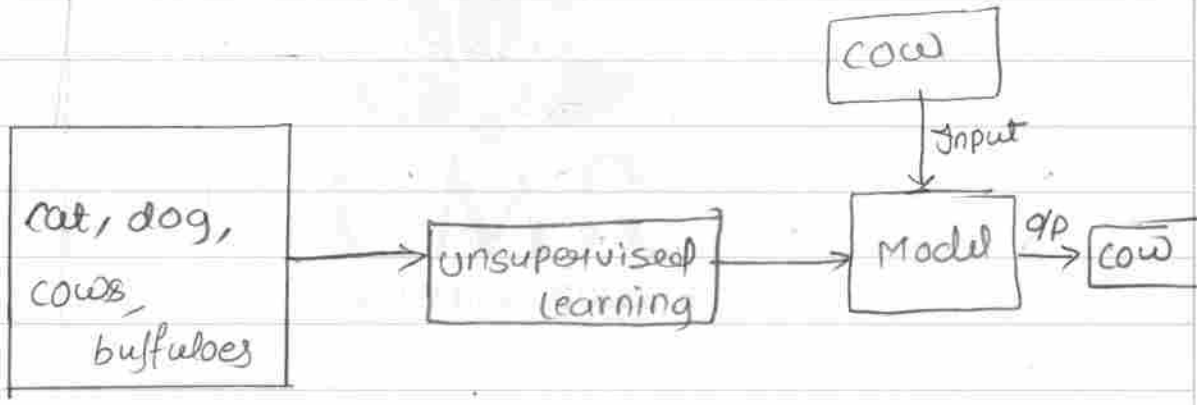
ii) unsupervised learning:-

we are not giving the labelled data for training. when we give the unlabelled data for training we are training the model using unsupervised learning technique. for example:- let us consider the 'Recommendation System' where initially we don't have the data of user. So it doesn't show the recommendation. Once, the user starts browsing the web, it keeps on tracing it which is unlabelled data & then recommendation comes based on the searching pattern.



iii) Reinforcement learning:-

Reinforcement learning means learning from mistake. for example:- let us suppose we have a Robot, so, there are two moves 1) step to the right option 2) step to the fire place. if the the Robot steps to the fire place he will get the negative ~~positive~~ rewarding. So next time Robot will remember not to step in fire. in the way of Rewards & panalty manner the training method is offered as Reinforcement learning.

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11. Social Media Analytics:-

Analysing the social trends such as most liked products, most commuted products. will helps the sales person to earn money. So, social media Analytics means analysing the social media produced data & getting a visualization to help the end user.

Feature/Advantages of Social Media Analytics:-

→ helps to learn the Market.

→ understand the human/end-user mentality.

→ helps to understand the future/present needs.

→ helps in generating revenue.

→ to understand the Demand & Supply

for example:-

let us take Amazon, it will use the social Media Analysis in way that it will help the Amazon to learn the most buyed product, most replaced product. so, that Amazon can understant the demand of perticular product.

Importance of social Media Analytics:-

→ it helps the vendor to understand the demand of perticular product & helps to earn the revenue.

→ helps in generating the revenue.

→ helps in visudizing the demand in easy way.

→ understance the present secenario.

→ If the vendor understand the social media Analytics He will be able to grow & learn from the market.

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if the vendor understand the above graph he will get to know the effect of price on demand. as a result he will have a proper mindset of the people.

→ visualization helps in the understanding the data clearly.

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Part-A

1) the input & outputs of Map Reduce:

the input of the MapReduce is the flume, sqoop collected & stored data. flume collects the unstructured data. & sqoop collects the structured data.

the output of MapReduce is the list, json form. which is given to the cloudera to plot or visualize the output of MapReduce.

2) Data moved in & out of Hadoop?

the input of the hadoop is the three types of data i) structured ii) unstructured iii) semi-structured. these input is taken by the flume & sqoop. that is stored in HBase, HDFS.

the output is produce in the cloudera to the end user.

3) secondary Name Node :-

→ it main function of secondary Name Node is to generate / store the logs. If the system fails & then the user will get helped with the log generated every minute.

→ the log is stored in the file called 'fsimage'.

→ it is the global master node.

→ logs are generated every minute in order to avoid the crashes.

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4) Supervised learning:-

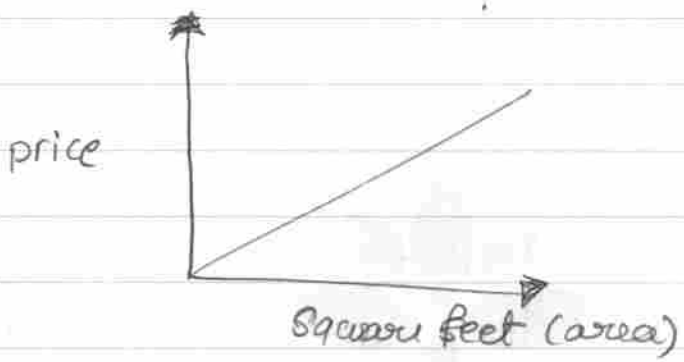
when/while training the data we are giving the labelled data this method of training is referred to as supervised learning.

→ technique involves :-

i) linear Regression

ii) classification

iv) Kmean algorithm.



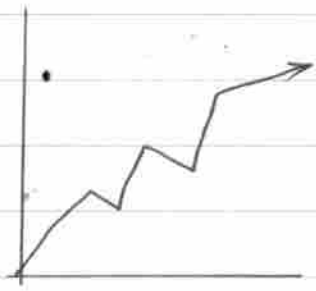
linear Regression

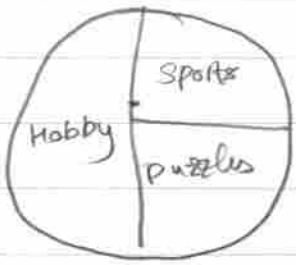
5) i) Data Analytics using R.

ii) visualization in R.

these are the application of R in Data Analytics

visualization means converting of raw data into the form of box-plot, histogram, pie charts etc.





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